KAIZEN MATHS

Lesson starter

Upper and Lower Bounds

Bounds in Formulae

Name _____

S1

Find the required bound for each geometrical quantity.

A rectangle has length 10.3 cm to the nearest 0.1 and width 11.1 cm to the nearest 0.1. Find the upper bound for its area.

S2

Find the required bound for each compound measure.

A journey distance is 312 m to the nearest whole number and the time is 22.3 s to the nearest 0.1. Find the upper bound for the speed.

S3

Use the given formula and rounded values to find the required bound.

A cylinder has radius 10.5 cm to the nearest 0.1 and height 14.4 cm to the nearest 0.1. Using $V = 3.142r^2h$, find the upper bound for V.

S4

Find the required bound for each geometrical quantity.

A rectangle has length 10.4 cm to the nearest 0.1 and width 5.5 cm to the nearest 0.1. Find the lower bound for its perimeter.

Show your working

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A journey distance is 312 m to the nearest whole number and the time is 22.3 s to the nearest 0.1. Find the upper bound for the speed.

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S4

Find the required bound for each geometrical quantity.

A rectangle has length 10.4 cm to the nearest 0.1 and width 5.5 cm to the nearest 0.1. Find the lower bound for its perimeter.

NUM-E608-L3 · S

Show your working

KAIZEN MATHS

Exit ticket

Upper and Lower Bounds

Bounds in Formulae

Name _____

EA1

Find the required bound for each compound measure.

A journey distance is 446 m to the nearest whole number and the time is 75.6 s to the nearest 0.1. Find the lower bound for the speed.

EA2

Use the given formula and rounded values to find the required bound.

A cylinder has radius 11.6 cm to the nearest 0.1 and height 18.5 cm to the nearest 0.1. Using $V = 3.142r^2h$, find the lower bound for V.

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KAIZEN MATHS

Exit ticket

Upper and Lower Bounds

Bounds in Formulae

Name _____

EB1

Find the required bound for each geometrical quantity.

A rectangle has length 21.5 cm to the nearest 0.1 and width 10.5 cm to the nearest 0.1. Find the lower bound for its perimeter.

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KAIZEN MATHS

Exit ticket

Upper and Lower Bounds

Bounds in Formulae

Name _____

EA1

Find the required bound for each compound measure.

A journey distance is 446 m to the nearest whole number and the time is 75.6 s to the nearest 0.1. Find the lower bound for the speed.

EA2

Use the given formula and rounded values to find the required bound.

A cylinder has radius 11.6 cm to the nearest 0.1 and height 18.5 cm to the nearest 0.1. Using $V = 3.142r^2h$, find the lower bound for V.

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KAIZEN MATHS

Exit ticket

Upper and Lower Bounds

Bounds in Formulae

Name _____

EB1

Find the required bound for each geometrical quantity.

A rectangle has length 21.5 cm to the nearest 0.1 and width 10.5 cm to the nearest 0.1. Find the lower bound for its perimeter.

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Show your working

Teacher answers & working

Upper and Lower Bounds · Bounds in Formulae

NUM-E608-L3 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S1

Find the required bound for each geometrical quantity.

A rectangle has length 10.3 cm to the nearest 0.1 and width 11.1 cm to the nearest 0.1. Find the upper bound for its area.

Answer:

$115.403\ \text{cm}^2$

Working:

Write the error interval for l: 10.25 cm ≤ l < 10.35 cm.

Write the error interval for w: 11.05 cm ≤ w < 11.15 cm.

Area = l × w.

Use upper bounds because the result must be as large as possible.

Upper bound = 10.35 × 11.15 = 115.403.

S2

Find the required bound for each compound measure.

A journey distance is 312 m to the nearest whole number and the time is 22.3 s to the nearest 0.1. Find the upper bound for the speed.

Answer:

$14.045\ \text{m/s}$

Working:

Speed = $\frac{distance}{time}$.

Distance interval: 311.5 m ≤ d < 312.5 m.

Time interval: 22.25 s ≤ t < 22.35 s.

For maximum speed, use the upper bound of distance and the lower bound of time.

Upper bound = $\frac{312.5}{22.25}$ = 14.045 m/s.

Teacher answers & working

Upper and Lower Bounds · Bounds in Formulae

NUM-E608-L3 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S3

Use the given formula and rounded values to find the required bound.

A cylinder has radius 10.5 cm to the nearest 0.1 and height 14.4 cm to the nearest 0.1. Using $V = 3.142r^2h$, find the upper bound for V.

Answer:

5053.34 cm^3

Working:

Radius interval: 10.45 cm ≤ r < 10.55 cm.

Height interval: 14.35 cm ≤ h < 14.45 cm.

$V = 3.142r^2h$. Since r and h are positive, use upper bounds for the upper bound of V.

Upper bound = $3.142 \times 10.55^2 \times 14.45$.

Upper bound = 5053.34 cm^3.

S4

Find the required bound for each geometrical quantity.

A rectangle has length 10.4 cm to the nearest 0.1 and width 5.5 cm to the nearest 0.1. Find the lower bound for its perimeter.

Answer:

31.6 cm

Working:

Write the error interval for l: 10.35 cm ≤ l < 10.45 cm.

Write the error interval for w: 5.45 cm ≤ w < 5.55 cm.

Perimeter = 2(l + w).

Use lower bounds because the result must be as small as possible.

Lower bound = 2(10.35 + 5.45) = 31.6.

Teacher answers & working

Upper and Lower Bounds · Bounds in Formulae

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EA1

Find the required bound for each compound measure.

A journey distance is 446 m to the nearest whole number and the time is 75.6 s to the nearest 0.1. Find the lower bound for the speed.

Answer:

$5.889\ \text{m/s}$

Working:

Speed = $\frac{distance}{time}$.

Distance interval: 445.5 m ≤ d < 446.5 m.

Time interval: 75.55 s ≤ t < 75.65 s.

For minimum speed, use the lower bound of distance and the upper bound of time.

Lower bound = $\frac{445.5}{75.65}$ = 5.889 m/s.

EA2

Use the given formula and rounded values to find the required bound.

A cylinder has radius 11.6 cm to the nearest 0.1 and height 18.5 cm to the nearest 0.1. Using $V = 3.142r^2h$, find the lower bound for V.

Answer:

$7733.33\ \text{cm}^3$

Working:

Radius interval: 11.55 cm ≤ r < 11.65 cm.

Height interval: 18.45 cm ≤ h < 18.55 cm.

$V = 3.142r^2h$. Since r and h are positive, use lower bounds for the lower bound of V.

Lower bound = $3.142 \times 11.55^2 \times 18.45$.

Lower bound = $7733.33\ \text{cm}^3$.

Teacher answers & working

Upper and Lower Bounds · Bounds in Formulae

NUM-E608-L3 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

EB1

Find the required bound for each geometrical quantity.

A rectangle has length 21.5 cm to the nearest 0.1 and width 10.5 cm to the nearest 0.1. Find the lower bound for its perimeter.

Answer:

63.8 cm

Working:

Write the error interval for l: 21.45 cm ≤ l < 21.55 cm.

Write the error interval for w: 10.45 cm ≤ w < 10.55 cm.

Perimeter = 2(l + w).

Use lower bounds because the result must be as small as possible.

Lower bound = 2(21.45 + 10.45) = 63.8.